Assignment 1: Kinematics and Perception

Due: 23:59, Oct. 12, 2026 | 4 questions | 100 marks

Full Name: _____

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If you have any questions about this assignment or its grading, please contact **ZHANG Zhen** at `zzhang2@mae.cuhk.edu.hk`.

Guidelines:

- Write your solutions with clear steps on the answer sheet, blank paper, or single-lined paper.
- Only handwritten work produced with a physical or digital pen is accepted.
- Write your full name, student ID, and page number on every answer sheet.
- Scan all answer sheets clearly, save them as one **single PDF file**, and upload it to Blackboard.
- Late submission policy applies.
- Only PDF files will be accepted.
- Keep the original hard copy after submission in case it is needed for checking.
- You may use a mobile scanning app such as CamScanner or Microsoft Lens.

QUESTION 1: Differential-drive kinematics — 25 MARKS

A no-slip differential-drive robot has reference point P midway between its wheels. Each wheel has radius $r = 0.10\,\text{m}$ and lies $l = 0.25\,\text{m}$ from P (track width $2l$). At one instant,

$$\theta = \frac{\pi}{3}, \qquad \dot{\varphi}_1 = 6.00\,\text{rad/s}, \qquad \dot{\varphi}_2 = 2.00\,\text{rad/s}.$$

Use the lecture's wheel numbering: positive rates roll forward and $\dot{\theta} = r(\dot{\varphi}_1 - \dot{\varphi}_2)/(2l)$. The local X_R-axis points forward, and $R(\theta)$ maps global velocity to the robot frame:

$$\dot{\xi}_R = R(\theta)\dot{\xi}_I, \qquad \dot{\xi}_I = \begin{bmatrix} \dot{x} & \dot{y} & \dot{\theta} \end{bmatrix}^T.$$

(a) Robot-frame velocity [7 marks]. Find $\dot{x}_R$, $\dot{y}_R$, and $\dot{\theta}$. Explain your value of $\dot{y}_R$.

(b) Global-frame velocity [8 marks]. Evaluate $R(\theta)$ and $R^{-1}(\theta)$, then find $\dot{\xi}_I$.

(c) Inverse calculation [10 marks]. At the same heading, a new instantaneous command is specified in the global frame as

$$\dot{\xi}_I^{\text{des}} = \begin{bmatrix} 0.30\,\text{m/s} \\ 0.30\sqrt{3}\,\text{m/s} \\ -0.40\,\text{rad/s} \end{bmatrix}.$$

Transform it to the robot frame, check the no-lateral-slip condition, and find the required wheel rates $\dot{\varphi}_1$ and $\dot{\varphi}_2$ from the forward model.

QUESTION 2: Fixed-wheel constraints and maneuverability — 25 MARKS

Use the fixed standard wheel model from the lecture:

$$\begin{bmatrix}\sin(\alpha+\beta) & -\cos(\alpha+\beta) & -l\cos\beta\end{bmatrix} R(\theta)\dot{\xi}_I - r\dot{\varphi} = 0$$

$$\begin{bmatrix}\cos(\alpha+\beta) & \sin(\alpha+\beta) & l\sin\beta\end{bmatrix} R(\theta)\dot{\xi}_I = 0.$$

The equations are the rolling and sliding constraints, respectively. Let $\theta = 0$, so the robot and global axes coincide. Both wheels below are fixed, have $r = 0.10\,\text{m}$, and impose the only support-wheel constraints considered.

Initially, consider only wheel A, with

$$l_A = 0.30\,\text{m}, \qquad \alpha_A = 0, \qquad \beta_A = \frac{\pi}{2}.$$

Its robot-frame contact point is $(0.30, 0)$, and it rolls along $+X_R$.

(a) Single-wheel constraints [7 marks]. Substitute wheel A's parameters. Give the numerical rolling and sliding rows, both scalar constraints, and $R(0)$.

(b) Admissible motion for wheel A [5 marks]. Test whether the candidate chassis velocity

$$\dot{\xi}_I = \begin{bmatrix} 0.40\,\text{m/s} \\ -0.15\,\text{m/s} \\ 0.50\,\text{rad/s} \end{bmatrix}$$

satisfies wheel A's sliding constraint. Show the residual. If admissible, find $\dot{\varphi}_A$ from the rolling constraint.

Now attach wheel B rigidly to the same chassis, with

$$l_B = 0.30\,\text{m}, \qquad \alpha_B = \pi, \qquad \beta_B = -\frac{\pi}{2}.$$

Wheel B is at $(-0.30, 0)$ and also rolls along $+X_R$. The contact points therefore lie on the longitudinal axis, not a common transverse axle.

(c) Complete-robot sliding constraints [8 marks]. Derive wheel B's sliding row, stack both rows into C_1, and find its rank. Test whether the velocity in (b) remains admissible.

(d) Maneuverability [5 marks]. Find the mobility δ_m, steerability δ_s, and maneuverability δ_M. State the allowable instantaneous velocities and whether no-slip rotation in place is possible.

QUESTION 3: From world coordinates to image pixels — 25 MARKS

An ideal pinhole camera has intrinsic matrix K and world-to-camera homogeneous transform E:

$$K = \begin{bmatrix} 800 & 0 & 320 \\ 0 & 800 & 240 \\ 0 & 0 & 1 \end{bmatrix}, \qquad E = \begin{bmatrix} 0 & -1 & 0 & 0.20 \\ 1 & 0 & 0 & -0.10 \\ 0 & 0 & 1 & 2.00 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Entries of K are in pixels; 3D coordinates and translations are in metres. Ignore distortion. Use a front virtual image plane, $Z_C > 0$, and

$$P_C = RP_W + t, \qquad \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & t \end{bmatrix} \begin{bmatrix} P_W \\ 1 \end{bmatrix}.$$

Consider the world point

$$P_W = \begin{bmatrix} 0.40 \\ 0.10 \\ 0 \end{bmatrix} \text{ m}.$$

(a) Identify the parameters [5 marks]. Extract R and t from E, and identify α_u, α_v, u_0, and v_0. Distinguish the roles of K and E.

(b) Two-step projection [8 marks]. Find P_C, then use perspective division to find (u, v). Show both steps.

(c) Projection matrix [6 marks]. Compute $M = K[R \mid t]$, use it to verify (b), and give the homogeneous scale λ.

(d) Back-projection [6 marks]. For the pixel in (b), parameterize the corresponding camera-frame points by $Z_C > 0$. Find the world point at $Z_C = 4.00\,\text{m}$, and explain why one pixel does not determine a unique 3D point.

QUESTION 4: Stereo vision and linear triangulation **25 MARKS**

Two ideal pinhole cameras have the same orientation and intrinsic matrix

$$K_1 = K_2 = \begin{bmatrix} 600 & 0 & 320 \\ 0 & 600 & 240 \\ 0 & 0 & 1 \end{bmatrix}.$$

Their axes are parallel, and the world frame is the left camera frame. Their centers are

$$C_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}, \qquad C_2 = \begin{bmatrix} b \\ 0 \\ 0 \end{bmatrix}, \qquad b = 0.20\,\text{m}.$$

The same stationary 3D point appears at

$$(u_1, v_1) = (440, 300), \qquad (u_2, v_2) = (410, 300).$$

Treat these values as exact until (d), and ignore distortion and occlusion. Use $P_{C_i} = P_W - C_i$, so $M_i = [I \mid -C_i]$ in normalized coordinates and KM_i in pixels.

(a) Normalize and set up [8 marks]. Compute the homogeneous normalized image vectors

$$\widetilde{p}_i = K^{-1} \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix},$$

and write the numerical 3×4 matrices M_1 and M_2. Explain the sign of M_2's translation column.

(b) Linear triangulation [10 marks]. Eliminate the image scales. Write two equations per camera and assemble

$$A\widetilde{P}_W = 0, \qquad \widetilde{P}_W = \begin{bmatrix} X_W & Y_W & Z_W & 1 \end{bmatrix}^T.$$

Solve for P_W. Direct algebra is sufficient; no numerical SVD is required. Nonzero row scalings are acceptable.

(c) Independent checks [7 marks]. Find the disparity $d = u_1 - u_2$ and verify the depth with the stereo formula (use focal length in pixels). Reproject the point to verify both observations.

End of assignment. Total: 100 marks.